

Obstructions and Formalized Identities in Two Binary Goldbach Research Programmes

A Negative-Results and Formalization Report

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Abstract

We record obstructions and established formalizations in an analytic corpus and an algebraic certificate campaign concerning binary Goldbach. The analytic results include exact coefficient projection, finite character extraction, canonical logarithm quantization, finite-field roots, and credited portions of the Mellin inversion dependency chain. Their scope is separated from uncompiled truncation sources and from the signed estimate still required for a prime-pair lower bound. The algebraic results include ordinary-polynomial soundness, calibration, uniform common-zero constructions for specified constraint families, and a rational static Nullstellensatz degree lower bound. Finite measurements attain that bound at the measured points; a universal matching construction is not established. Polynomial-calculus measurements and its explicitly assumed external lower-bound predicate are kept distinct. A completed integer and CRT computation at a single finite input is reported without a Lean refinement claim. An evidence map and generated Lean inventory identify the supporting artifacts and their exact verification levels. The report establishes no conclusion resolving binary Goldbach and excludes neither the full analytic programme nor the full algebraic certificate route.

MSC2020: 11P32 (primary); 68V20, 03F20, 13P10 (secondary).^[R-MS]

Suggested arXiv classification: math.NT; possible cross-lists cs.LO and cs.CC, subject to the author’s submission decision.

1 Introduction and scope

Binary Goldbach asks whether every even integer greater than two is the sum of two primes.^[R-FROZEN] The distinction between a useful identity and a sufficient prime-pair lower bound is central to this report. The classical context includes the Hardy–Littlewood formulation, Vinogradov’s three-prime theorem, Chen’s prime and almost-prime theorem, and the ternary theorem of Helfgott.^[1, 2, 3, 4] The published finite verification up to $4 \cdot 10^{18}$ is a different kind of evidence from a theorem covering all even integers.^[5]

We consolidate the round-22 analytic corpus as summarized in Synthesis V3.1 and the stopped AlgebraicGoldbach campaign.^[R-CORPORA] Existing material from the historical Horizon repository is admitted only where an existing Judge record supplies the stated credit; a source file or a project README alone is insufficient. No new mathematical execution was performed

for this manuscript.[R-STOP] Classical projection, Mellin inversion and finite counting arguments are recorded or formalized here without claims of novelty. Literature on Nullstellensatz and polynomial calculus supplies context, rather than a replacement for the hypotheses of the project statements. [6, 7, 8, 9, 10]

How to read this report.

First read the evidence label attached to a result.

Then read its domain, hypotheses and mathematical object.

Finally consult its evidence key in the machine-readable map and the inventory before treating a source or finite observation as a theorem.

2 Evidence levels and verification protocol

Level	Meaning in this report
Compiled Lean	An existing successful Lean record covers the declaration; the available axiom record is reported. It does not certify a different informal statement or remove its explicit hypotheses.
SOURCE	A written Lean source exists without the required successful compile credit for that item.
PAPER	An existing written mathematical argument or review is recorded; it is not promoted to compiled formal proof.
Observed	An existing process, file or finite measurement was recorded; the observation is limited to that run and input.
EVIDENCE	Existing independently checked finite arithmetic, model or certificate data support precisely the stated finite claim.

These levels do not form an automatic implication chain. In particular, source availability, a passing build, an axiom list and a finite certificate answer different questions. Explicit theorem premises remain premises even when the declaration uses only the standard logical axioms.[R-PROTOCOL]

The campaign protocol used separate implementation, adversarial reading and Judge roles, source pins, exact ordinary polynomial definitions, and valid-data-first corruption controls. The accepted evidence identifies the actual command and result, rather than treating an agent’s summary as a proof.[R-PROTOCOL] The paper’s fresh compilation and claim-to-evidence review concern this document only. They neither rerun the mathematical experiments nor add Judge credit to their results.[R-STOP]

2.1 Limitations of the verification arrangement

The agent roles share a model provider and are not an external mathematical referee panel. Their separation is operational; it does not imply statistical independence or third-party validation. The Lean kernel checks formal terms under their recorded assumptions. Interpretation of those terms, choice of constraints, and correspondence with a native program still require human review.[R-PROTOCOL]

The preserved Windows artifacts distinguish raw bytes, Git blobs and historical documentary snapshots. CRLF transport and path separators caused reader and binding problems; named corrections do not justify silently normalizing old reports. The manuscript cites the

original artifact hashes and keeps documentary transport distinct from mathematics.^[R-WINDOWS] The repository already contains the LF text policy requested for this publication; no existing file was converted for the paper.^[R-ATTRIBUTES]

The algebraic frozen target is a bare, unfinished theorem declaration. It contains no proof term and is excluded from the auxiliary build. Freezing the entire file’s bytes prevents adding a proof without changing the file hash. Thus the frozen-file rule cannot simultaneously accept a newly added proof of that declaration. This is a design flaw in the target acceptance contract, not a mathematical obstruction. The frozen bytes remain unchanged.^[R-FROZEN]

3 Analytic framework and the credited dependency chain

The analytic corpus supplies exact identities and explicit approximation bounds. Its recorded Judge receipts distinguish successful module rows from the status of an entire batch. This distinction matters for batches 6, 7, 8, 9, 11 and 26: each globally failed batch retains only the whole-module PASS rows identified in the inventory. No failed or uninvoked module is credited. The standard axioms recorded for credited declarations are subsets of **propext**, **Classical.choice** and **Quot.sound**. Appendix A.1 provides a module table; the accompanying JSON and CSV retain every extracted axiom-audit row.^[A-PARTIAL-BATCHES] ^[A-INVENTORY-COUNTS]

3.1 The genuine coefficient and its continuous projection

Throughout this section, Λ is the genuine von Mangoldt function: $\Lambda(0) = \Lambda(1) = 0$, $\Lambda(p^k) = \log p$ for a prime p and an integer $k \geq 1$, and it vanishes otherwise. For $a > 0$, set

$$T_a(\theta) = \sum_{n \geq 0} \Lambda(n) e^{-an} e^{in\theta}, \quad C_N = \sum_{n=0}^N \Lambda(n) \Lambda(N-n). \quad (1)$$

The series is absolutely convergent. The compiled continuous projection identity, for every natural N , is

$$C_N = \frac{e^{aN}}{2\pi} \int_0^{2\pi} T_a(\theta)^2 e^{-iN\theta} d\theta. \quad (2)$$

This is **thermalContinuousProjection_eq_vonMangoldt_sum** in the module **ThermalProjectionIdentity22**. The source uses the full interval $[0, 2\pi]$. Prime powers remain in C_N ; the equality alone is not a lower bound for a prime-pair sum. The square T_a^2 extracts additive pairs. Replacing it by $|T_a|^2$ extracts differences and changes the coefficient being studied.^[A-CIRCLE] ^[A-OBSTRUCTION-SIGNED-GAP]

There is also a compiled elementary truncation envelope. With $r = e^{-a}$, let

$$U(a) = \frac{r}{(1-r)^2}, \quad B(a, M) = r^{M+1} \left(\frac{M+1}{1-r} + U(a) \right). \quad (3)$$

If $T_{a,M}$ is the sum through M , the corresponding normalized projection differs from the full projection by at most $2e^{aN}U(a)B(a, M)$ in complex norm. The bound follows from actual von Mangoldt terms and a constructed summable majorant, rather than from an assumed coefficient bound. Its module is a retained PASS row in failed batch 11.^[A-CIRCLE]

3.2 Finite projection and canonical logarithm quantization

For natural $N \leq M$ and $K > \max(N, 2M - N)$, the discrete projector satisfies

$$C_N = \frac{e^{aN}}{K} \sum_{j=0}^{K-1} T_{a,M}(2\pi j/K)^2 e^{-2\pi i N j/K}. \quad (4)$$

The condition excludes all unwanted exponents congruent to N modulo K . This is a finite identity for every real a , including $a = 0$; it does not assert convergence of the infinite series at $a = 0$. The declaration `samplerTrueCircle_eq_coefficient` in `DiscreteThermalProjectio` `n22` records the concrete angular grid.[A-DISCRETE]

The quantized construction uses the fixed scale $S = 2^{58}$. Define

$$L_{32}(z) = 2 \sum_{j=0}^{31} \frac{z^{2j+1}}{2j+1}, \quad \rho_{32} = \frac{9}{4 \cdot 65 \cdot 3^{65}}. \quad (5)$$

For $2 \leq p \leq 10^8$, write $k = \lfloor \log_2 p \rfloor$ and $z = (p - 2^k)/(p + 2^k)$. The rational lower endpoint is $b_p = kL_{32}(1/3) + L_{32}(z)$, with interval width $(k+1)\rho_{32}$. The constructor rounds the midpoint multiplied by S to the nearest integer, resolving exact ties toward the even integer, then clamps to $[0, 32S]$. The proved error is at most $1/S$; precision is a conclusion of this fixed constructor.[A-LOG]

Applying the constructor to the base prime of each prime power defines integers A_n and $q_n = A_n/S$. Both q_n and $\Lambda(n)$ lie in $[0, 32]$ on $0 \leq n \leq 10^8$, and $|q_n - \Lambda(n)| \leq 1/S$ there. Thus, for natural $N \leq 10^8$, the canonical integer coefficient $I_N = \sum_{n=0}^N A_n A_{N-n}$ satisfies

$$\left| C_N - \frac{I_N}{S^2} \right| \leq E_{\log}(N) := (N+1) \left(\frac{64}{S} + \frac{1}{S^2} \right). \quad (6)$$

The fixed integer guard proves $2E_{\log}(10^8) \leq 10^{-6}$. These statements are in `RationalLogQ` `uantization22` and `QuantizedLambdaEnvelope22`; no real-logarithm precision assumption is inserted.[A-LOG]

3.3 Finite-field extraction and the five concrete roots

The generic finite-field module derives character orthogonality from an actual primitive root, then instantiates the coefficient projector for the canonical integer sequence. The concrete root module proves the primality of the five moduli and exact order $K = 2^{27}$ of the roots $\omega = g^{(p-1)/K}$ in their respective fields. Here g denotes the specified bank base; the needed assertion is the order of ω . [A-FIELD]

Certified prime p	Bank base g
2013265921	31
2281701377	3
3221225473	5
3489660929	3
3892314113	3

At $N = M = 10^8$, each such field projector returns I_N modulo p . The compiled results are in `FiniteFieldProjection22`, `ConcreteNTTRoots22` and `ConcreteNTTA32Projection22`. A theorem about this finite projector does not by itself refine the native transform or CRT program. The saved numerical computation is reported separately in Section 7.[\[A-FIELD\]](#)

3.4 The actual Gamma inversion dependency chain

The inversion chain is a sequence of credited source modules with explicit imports, rather than an assumed Gamma inversion formula. The closed-strip prerequisite bounds the actual Gamma function by $2\exp(-\pi|\operatorname{Im} s|/4)$ for $1 \leq \operatorname{Re} s \leq 2$. The positive-real inverse is first proved; the local complex module constructs its branch and integrable majorants; the holomorphy module then extends the identity to the right half-plane. [\[A-GAMMA-CHAIN\]](#)

Module	Batch; rows	Retained scope
<code>GammaPrerequisites22</code>	2; 23	Actual rotation and strip bound
<code>ThermalGammaMellinInverse22</code>	20; 11	Positive-real inversion
<code>ComplexGammaMellinLocal22</code>	26; 22	Local bounds and real agreement; partial PASS
<code>ComplexGammaMellinHolomorphy22</code>	27; 15	Complex inversion for $\operatorname{Re} w > 0$
<code>ComplexGammaMellinLambda22</code>	35; 30	Actual infinite Λ interchange
<code>ComplexGammaMellinTail22</code>	30; 22	Unweighted Gamma tail
<code>ComplexGammaMellinExpTailBridge22</code>	32; 2	Exponential inversion error

The first five rows follow the import chain in their displayed order. Batch 26 failed globally, but its entire local inversion module has the retained PASS row. The unweighted tail module imports the local module; the exponential-tail bridge imports both holomorphy and tail. The Λ interchange imports holomorphy. It does not require the uncompiled weighted- Λ tail module. Counts in this table are saved axiom-audit declaration rows, including definitions.[\[A-GAMMA-CHAIN\]](#)

3.5 The genuine infinite sum–integral interchange

For real t and complex w with $\operatorname{Re} w > 0$, set

$$D(t) = \sum_{n \geq 2} \Lambda(n) n^{-2-it}, \quad P(w) = \sum_{n \geq 0} \Lambda(n) e^{-nw}. \quad (7)$$

The source defines these actual arithmetic series. It proves continuity of D , the bound $|D(t)| \leq 6$, and summability of the integrals of norms needed for the infinite interchange. In particular, the coefficient majorant is bounded by $2n^{-3/2}$ and its total mass by 6. The compiled conclusion is

$$P(w) = \frac{1}{2\pi} \int_{\mathbb{R}} D(t) \Gamma(2+it) w^{-2-it} dt, \quad \operatorname{Re} w > 0, \quad (8)$$

with principal complex powers. The declaration `lambdaMellinThermal_eq_integral` in `ComplexGammaMellinLambda22` is therefore an infinite arithmetic interchange result. This module alone does not identify D with a zeta logarithmic derivative or supply a weighted- Λ tail bound uniform on the coefficient contour. [\[A-MELLIN\]](#)

3.6 Angular integration by parts and the remaining boundary

For $a > 0$, natural $N \geq 1$ and $q \in \mathbb{C}$, the compiled angular module uses

$$J_{a,N}(q) = \frac{1}{2\pi} \int_{-\pi}^{\pi} (a - i\theta)^{-q} e^{-iN\theta} d\theta. \quad (9)$$

Its exact recurrence is

$$J_{a,N}(q) = B_{a,N}(q) + \frac{q}{N} J_{a,N}(q+1), \quad (10)$$

$$B_{a,N}(q) = \frac{i(-1)^N}{2\pi N} ((a - i\pi)^{-q} - (a + i\pi)^{-q}). \quad (11)$$

The boundary term at $q = 1$ is proved nonzero. Periodicity of the complete arithmetic trace therefore cannot be substituted for periodicity of an individual principal-branch Mellin kernel. This gives a precise boundary obstruction to dropping that term; it does not rule out estimates that retain it. [A-IBP] [A-OBSTRUCTION-BOUNDARY]

The unweighted Gamma-tail and exponential-inversion-error theorems are compiled. The repaired weighted- Λ tail, uniform circle geometry, coefficient truncation envelope, scalar connectors and coefficient-to-scalar bridge are retained source only. Failed batch 36 adds no credit: the weighted-tail module failed, and the geometry module was not invoked. Consequently the complete Mellin-to-coefficient truncation chain remains uncompiled. A norm enclosure, even if supplied by that chain, would still leave the signed estimate and prime-power corrections needed for a prime-pair conclusion. [A-TRUNCATION-BOUNDARY] [A-OBSTRUCTION-SIGNED-GAP]

4 Algebraic certificate setting and family obstructions

We record an auxiliary formalization in the ordinary polynomial ring $\mathbb{Q}[X_0, \dots, X_N]$. Let a_i be the prime indicator and retain the original ordered polynomial, including a square term at the midpoint,

$$g_N(X) = \sum_{i=0}^N X_i X_{N-i}.$$

A supplied certificate has the literal form

$$1 = \sum_j A_j F_j + \sum_{i=0}^N U_i (X_i^2 - X_i) + B g_N.$$

The compiled soundness theorem excludes a common zero of these actual generators. Applied to the prime point it still requires both the identity and genuine prime truth of the supplied family. It constructs neither. This is a certificate meta-theorem, not a proof of binary Goldbach [ALG-SETUP]

4.1 Calibration and circularity

The pinned control $F_i = X_i - a_i$ has an identity exactly when the ordered prime count is positive. Its explicit multipliers have degree at most one; their ordinary multiplier-generator summands

have degree at most two. These are different conventions. For the calibration D_m , consisting of Booleanity, nonprime zeros and total selected count $m \in \mathbb{N}$, the compiled C1 equivalence is

$$\exists x (D_m(x) \wedge g_N(x) = 0) \iff m \leq \pi(N) - r(N),$$

where $r(N)$ counts unordered prime representations, including a central loop. At $m = \pi(N)$ the family pins the prime vector. Thus exclusion of its pseudo-model is equivalent to positive prime representation count; it supplies a circular calibration control.^[ALG-C0-C1]

4.2 Family evidence and coordinate guards

Every original family has recorded Boolean pseudo-models at each mandatory test value $N \in \{30, 60, 100, 200\}$. The first failure *among that quartet* is therefore 30. The “first even” column below has a different, explicit scope: even $N \geq 4$, with every earlier even case covered by saved finite data. Its existing saved-data audit passed and its serialized result was reviewed before consolidation, but no scoped Judge/Trophy promotion was completed. Those entries are EVIDENCE, not new formal credit. No statement about smaller or odd N is implied.

Family	Uniform	First even	Coordinate guard	Models without g , $N = 30$
Coarse parity ^[ALG-FAMILY-Y-COARSE_PARITY]	Yes	4	No additional prime-coordinate pin	32768
SIEVE only ^[ALG-FAMILY-Y-SIEVE_ONLY]	Yes	4	Nonprime zeros; no prime-coordinate pin	1024
SIEVE + dyadic intervals ^[ALG-FAMILY-SIEVE_DYADIC_BERTRAND]	Yes	8	Direct $X_2 - 1$ pin	144
Full Bertrand coarse ^[ALG-FAMILY-BERTRAND_COARSE]	Yes	16	$m = 1$ interval pins X_2	2605
Full SIEVE ^[ALG-FAMILY-FULL_BERTRAND_SIEVE]	Yes	20	$m = 1$ interval pins X_2	40
Bertrand SIEVE + factors ^[ALG-FAMILY-FACTORS_COVERAGE_BERTRAND_SIEVE]	Yes	20	$m = 1$ X_2 pin; small-prime factor ≥ 2 pins	
Proper Bertrand only, $m \geq 2$ ^[ALG-FAMILY-PROPER_BERTRAND_ONLY]	Yes	4	Every coordinate free in Lean	1360168984

All seven rows have closed-form constants and more than one Boolean model without g at the quartet values; the displayed counts concern $N = 30$ only. All seven fail at every quartet value. Whole-vector multiplicity does not excuse an equation that fixes a prime coordinate. The dyadic family explicitly contains $X_2 - 1$; the $m = 1$ full Bertrand interval also fixes X_2 . Proper-divisor coverage separately fixes small prime coordinates. The proper interval-only family starts at $m = 2$, contains no coarse or SIEVE equations, and has compiled every-coordinate Boolean freedom. Its existing candidate credit stops at L5; no eligible family receives L6.^{[ALG-COORDINATE-GUARDS][ALG-NO-ELIGIBLE-FAMILY]}

The calibration and the later density control have a different status:

Control	Uniform	Coordinate status	Quartet status	First even $N \geq 4$
$D_{\pi(N)}$ [ALG-C0-C1]	No	Pins full prime vector	Exclusion iff $r(N) > 0$	Not a first-input survey
Coprime-divisor density[ALG-COPRIME-GATE]	Yes	Each coordinate free, all N	Recorded UNSAT at all four	4, Boolean common zero

The second control is rejected by its exact near-full-count classification; its quartet UNSAT records do not promote it to an eligible candidate.

4.3 Uniform and structural obstructions

The actual coarse/Bertrand and proper interval-only families have compiled rational Boolean common zeros with $g_N = 0$ for every $N \geq 24$. The coarse construction selects composite 9; it must not be silently used as a SIEVE model. Prime-supported finite witnesses are separate. [ALG-UNIFORM-BERTRAND] Proper-divisor coverage through $M \leq N^2$ is equivalent, over arbitrary rational points, to $X_p = 1$ for the prime coordinates with $p^2 \leq M$. The combined coarse/Bertrand family retains the explicit common zero under $N \geq 24$ and $M \leq (\lfloor N/2 \rfloor - 3)^2$. [ALG-FACTOR-HORIZON] Canonical rational equations whose coefficients all have one sign and which vanish at the prime point preserve every supplied prime-supported common zero. This statement does not cover arbitrary mixed-sign equations. [ALG-SIGN-DEFINITE]

The rational coprime-divisor family is exactly near-full prime selection: at most one prime coordinate may differ from one. Its literal quartet UNSAT data do not make it an eligible weaker family. [ALG-COPRIME-GATE] The actual k -ary arithmetic ideal equals the prime k -subset ideal, and its arbitrary rational models have fewer than k prime coordinates different from one. For every $k \geq 2$ the compiled $N = 4$ Boolean common zero excludes a certificate for that exact class. It is not a counterexample to Goldbach. [ALG-KARY-CLASSIFICATION]

Finally, on the actual complement-pair Boolean cube, a literal-product family has a common zero when its exact total clause weight is strictly smaller than $2^{\dim C_N}$, for even $N \geq 6$. The formalized dimension is the actual finite cube-variable cardinality; no closed floor formula is substituted for it. [ALG-LITERAL-CUBE] These statements exclude certificates for their declared classes and hypotheses only. They do not exclude the whole algebraic route. Exact theorem names and source hypotheses are catalogued in Appendix A.

5 Degree results and finite computation evidence

For the actual full-count calibration over $\mathbb{Q}$, the compiled lower theorem applies to every supplied ordinary certificate. Here A is the multiplier of the single count constraint $\sum_i X_i - \pi(N)$, not an arbitrary family multiplier A_j :

$$\deg A \geq \alpha(N), \quad \deg(A(\sum_i X_i - \pi(N))) \geq \alpha(N) + 1, \quad \alpha(N) = \pi(N) - r(N).$$

The required degree is unbounded along even N . Neither theorem asserts the existence of a certificate. Equality is established by existing exact finite arithmetic only at the 26 measured even points $N = 10, 12, \dots, 60$: the rational identities and the finite-field minima both equal $\alpha + 1$, with degrees between 3 and 14. This finite tightness is not a compiled universal upper theorem. The universal upper construction with midpoint loops remains blocked; its no-loop counterpart has only paper status. [ALG-NS-LOWER][ALG-NS-UPPER-LIMIT][ALG-DEGREE-CURVES]

Polynomial calculus is measured separately, with genuine ordinary powers, input rules, binary linear combinations and single-variable multiplication. The saved measurements over $\mathbb{F}_{1000000007}$ have exact degrees 3, 3, 3, 3, 4, 4, 4, 4, 4, 4, 5, 5 for even $N = 10, \dots, 30$; the rational measurements agree at those 11 points. The additional finite-field-only points $N = 32, 34, 36$ have degrees 6, 5, 5. All these computations are labeled EVIDENCE; they are not Lean computations, rational transfers from the larger points, or asymptotic bounds. The full degree rows are recorded in the publication JSON and Appendix A. [ALG-DEGREE-CURVES]

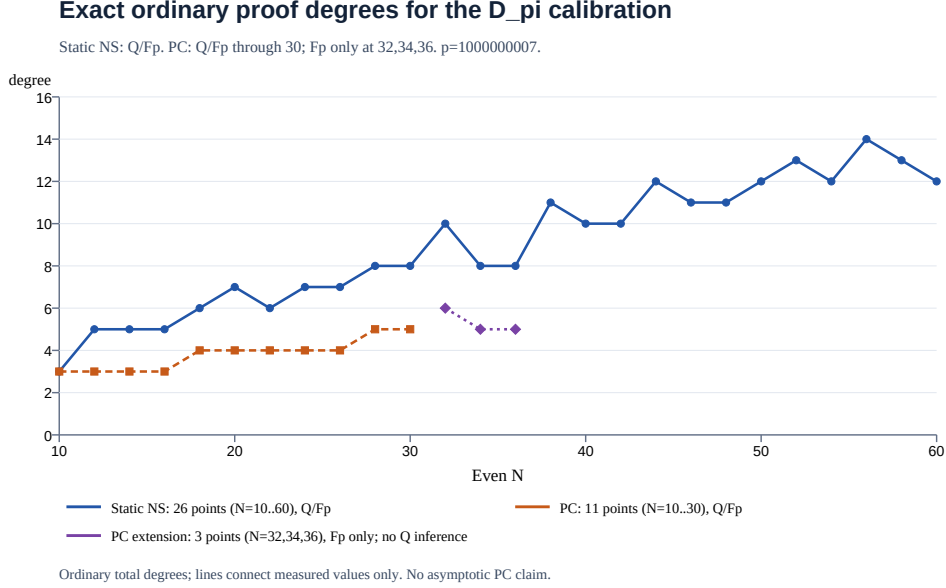


Figure 1: Existing finite degree plot. Static NS equality is measured over $\mathbb{Q}$ and $\mathbb{F}_{1000000007}$ at even $N = 10, \dots, 60$. PC degrees agree over both fields at even $N = 10, \dots, 30$; the points $N = 32, 34, 36$ are finite-field-only. The plot is evidence at these inputs, not a universal upper bound or a rational transfer.[ALG-DEGREE-FIGURE]

Grouping the existing CSV rows using α from the saved lower-bound JSON finds the same PC degree whenever α repeats in these 14 finite-field points. The complete observed groups are:

α	N	Observed PC degree
2	10	3
4	12, 14, 16	3
5	18, 22	4
6	20, 24, 26	4
7	28, 30, 34, 36	5
9	32	6

This is finite group consistency, not a universal dependence theorem or a closed PC degree formula.[ALG-PC-ALPHA-GROUPING]

The compiled ordinary-Q restriction sends a supplied D_π PC refutation to a genuine Boolean-knapsack refutation on α variables without increasing ordinary line degree. The explicit external

predicate `KnapsackLowerBound` remains a hypothesis of its lower corollary. That conditional corollary is out of scope and is recorded in Appendix B; the partial lattice, coefficient and count-action lemmas do not remove the hypothesis.[ALG-PC-HYPOTHESIS]

6 Obstruction catalogue

Table 1 collects the specific failed implications and restricted impossibility statements. Each row refers to the object defined in the corresponding section. A common zero for one family rules out a certificate for that family; it says nothing universal about all possible families. A failed sign argument leaves other analytic arguments open.

Table 1: Scope of the recorded obstructions.

Obstruction	Scope	Status	What it does not show
Exact projection lacks a signed lower bound	True additive coefficient	Compiled identities; open estimate	It does not supply a positive prime-pair margin.[R-ANALYTIC-GAP]
Absolute error control lacks cancellation	Quantization or truncation envelopes	Mixed compiled/SOURCE	A small approximation radius does not control the sign of the reference-minus-coefficient term.[R-ANALYTIC-GAP]
Boundary term retained	Individual Mellin kernel	Compiled Lean	Periodicity of the full trace does not delete each kernel boundary.[R-BOUNDARY]
Positive definite reconstruction lacks paired mass	The recorded restricted positive series	PAPER	Positivity alone does not imply a uniform strictly positive additive coefficient.[R-FEJER]
Negative direction of the reviewed kernel	Specific operator restriction	PAPER	It does not exclude every operator method.[R-KERNEL]
Cost of the proposed tail envelope	Particular written Mellin error envelope	PAPER / SOURCE	It is not a lower bound for the true error or for every evaluation algorithm.[A-OBSTRACTION-COST]
Finite arithmetic counterexamples and losses	The guarded block, mask, completion, rank and Type-II groups in Appendix A	Compiled Lean	They refute the specified intermediate claims, not the full analytic programme.[A-ARITHMETIC-OBSTRUCTIONS]
Circular or direct-pinning constraints	Calibration and named family equations	Compiled/SOURCE guard	They do not provide new information about the prime vector.[R-ALGEBRAIC]
Common zeros	Specified Bertrand, divisor and literal families	Compiled Lean / EVIDENCE	They do not exclude the complete certificate route.[R-ALGEBRAIC]
Growing certificate degree	Rational static full-count calibration	Compiled lower bound	It does not forbid unbounded degree or a different constraint family.[R-DEGREE]
Finite computation has finite scope	One completed integer/CRT input	EVIDENCE	It is not an all-input native refinement or an analytic threshold theorem.[R-NUMERICAL]

The sieve parity limitation is classical background, not a new theorem of this project.[11] The algebraic and analytic obstructions in this table have different mechanisms; no implication between them is asserted merely because each prevents a particular attempted argument.

7 The finite computation at $N = 10^8$

The saved numerical verdict concerns one input, $N = 100000000$, with transform length $K = 134217728 = 2^{27}$ and scale $S = 288230376151711744 = 2^{58}$. It records **EX-ACT_NUMERICAL_ARITHMETIC_VALIDATED** for the equality of the five-modulus CRT reconstruction, the direct canonical A32 integer sum and the auxiliary B40 integer sum. The documentary numerical check is separate from the compiled finite-field identities in Section 3. It is not a fresh execution of the transform or the prime catalogue.

[A-NUMERICAL]

Modulus	Saved residue
2013265921	577099750
2281701377	626366149
3221225473	232952773
3489660929	2618309304
3892314113	2671748987

The stored CRT product and reconstructed integer are, respectively,

$$P_{\text{CRT}} = 200988950952742247103310192629436327377560928257,$$

$$I_{10^8} = 14616354094337692277256326339388864358034646.$$

The same integer is recorded for the direct A32 and B40 sums, with actual difference zero. The saved catalogue contains 5761455 prime records. The coefficient denominator is

$$S^2 = 83076749736557242056487941267521536.$$

This denominator scales the quantized integer coefficient; the integer itself is not the real von Mangoldt coefficient.[A-NUMERICAL]

The saved logarithm-envelope numerators, over this denominator, are

$$E_{\log} : 1844674425817699235409551617,$$

$$2E_{\log} : 3689348851635398470819103234.$$

Their recorded decimal values are approximately

$$E_{\log} \approx 2.2204460714547736 \cdot 10^{-8},$$

$$2E_{\log} \approx 4.4408921429095471 \cdot 10^{-8}.$$

The joint radius is below 10^{-6} . The real-coefficient enclosure theorem in **QuantizedLambdaEnvelope22** applies to its mathematical canonical integer constructor. Connecting every native output to that constructor would require the native-program refinement that the verdict explicitly leaves unproved.[A-NUMERICAL][A-LOG]

The verdict also explicitly leaves B40 real-logarithm refinement and the Mellin coefficient-truncation chain unproved. It establishes neither the required signed D_N estimate nor a Goldbach conclusion. In particular, this single coefficient computation is not a verification of every even integer up to 10^8 , and equality of two saved integer sums does not make either unproved analytic refinement automatic.[A-NUMERICAL]

8 What a sufficient approach would still require

In the preserved analytic notation, C_N is the von Mangoldt convolution and includes proper prime powers, while R_{ref} is the reference quantity of the existing balance framework. A signed estimate for $R_{\text{ref}} - C_N$ remains necessary within that framework, together with the prime-power correction, boundary terms and its stated bridge conditions. A coefficient representation or enclosure does not supply that estimate.^[R-ANALYTIC-GAP] An ingredient sensitive to the prime versus almost-prime distinction is still required; invoking parity terminology alone provides no such input.

The analytic onset in the V3.1 framework is $\log N \geq 10^{24}$. The single finite computation at $N = 10^8$ is outside this regime. The published verification through $4 \cdot 10^{18}$ leaves the interval to that analytic onset uncovered by the present corpus. No theorem or computation in this report fills that interval.^{[R-THRESHOLD][5]}

For the algebraic route, a family would need genuinely sufficient prime-valid information without circularly assuming the desired nonvanishing, and an actual uniform identity with its ordinary degree accounted for. Lower bounds for a calibration do not substitute for that construction. The open matching static construction and the assumed polynomial-calculus lower-bound predicate remain separate obligations. ^{[R-DEGREE][R-PC]}

9 Reproducibility and archival scope

The publication repository is <https://github.com/seisner1967-hash/goldbach-horizon>. The pre-publication repository snapshot is `f7e1fda19a97`; the accepted algebraic proof snapshot is `e5c4e119473e`. Their recorded toolchain is Lean 4.15.0 with Mathlib revision `9837ca9d65d9`.^[R-REPRO] At the appropriate pinned source snapshot the project build command is

`lake build`

This manuscript preparation did not execute it. A successful auxiliary build does not prove the unfinished frozen target.^[R-FROZEN]

The generated inventory records theorem names, the available axiom lists, credit levels, source hashes and existing commit identifiers. Its scripts derive tables from existing artifacts only. The algebraic `AXIOMS.md` source and analytic audit records are identified in `evidence_map.json`. Appendix A explains their coverage and records omissions rather than inventing missing credit. ^[R-REPRO]

The arXiv source bundle contains portable TeX, the compiled bibliography and figures. Larger binary evidence is not included in this paper commit; its hashes and scope are recorded for a separate author-managed release. The source bundle does not need a local workspace path to compile.

A Generated Lean inventory

The tables and companion CSV/JSON inventories are generated from existing source declarations and audit artifacts. Definitions are identified as definitions; auxiliary declaration counts are not theorem counts. A declaration listed with SOURCE status is not listed as an established theorem. Explicit local hypotheses are retained in the informal statement. The table column “axioms” records what the available audit actually prints; an absent per-declaration audit is not filled by inference.

A.1 Analytic inventory from existing Judge artifacts

The electronic files `analytic_inventory.json` and `analytic_lean_inventory.csv` list every extracted axiom-audit row, its declaration kind, recorded axioms, textual declaration header, source and receipt paths, full hashes and existing repository commit. The header is extracted Lean text, not an elaborated type; surrounding section variables and hypotheses remain in the pinned source. A missing separate paraphrase is explicitly labelled. These tables are generated by `generate_analytic_inventory.py` from existing artifacts only. The recorded official counter is 88 modules and 1,488 mixed historical entries. The exhaustive axiom-row extraction gives 1,768 rows: 1,222 historical and 546 from round 22. The historical counter counts 942 theorems, while later axiom rows also contain definitions and structures. Consequently neither 1,488 nor 1,768 is presented as a theorem count. Definitions without an individual axiom row are not assigned an invented empty axiom list. The tables below select one recorded theorem per module and give a human paraphrase of its scope. They do not replace the original hypotheses in the pinned source. The electronic inventory retains all 1,768 rows. Commit `f7e1fda19a97` contains the existing public artifacts; source and receipt hashes identify their exact bytes. The axiom abbreviations are P=`propext`, C=`Classical.choice`, Q=`Quot.sound`. A partial label means module-row PASS in a globally FAILED batch. Batches 6, 7, 8, 9, 11 and 26 have such retained rows; failed or uninvoked rows are excluded.`[A-INVENTORY-COUNTS][A-PARTIAL-BATCHES]`

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
ParityWeights	<code>concrete_prime_triprime_same_parity</code> . The prime 101 and the composite $101 \cdot 103 \cdot 107$ have the same odd parity weight; the composite lies between 100 and 100000000 and has no prime divisor at most 100.	P,C,Q; Compiled Lean; <code>f7e1fda; 8c4954596ac6</code>
QuarticMobius	<code>weighted_boundary_identity_real</code> . For arguments between α and N with N at most α^4 , the weighted Mobius sum equals the stated combination of short double, cubic and quartic divisor sums.	P,C,Q; Compiled Lean; <code>f7e1fda; 1c370324b3d9</code>
ChenWeight	<code>weighted_prime_identity_real</code> . Under the quartic cap and the absence of prime factors at most α , the finite weighted Chen expression equals the corresponding prime-indicator sum.	P,C,Q; Compiled Lean; <code>f7e1fda; 65a38665aeae</code>
MultifibreObstruction	<code>literalFourSiteBlock_neg</code> . The explicitly defined four-site block is negative.	P,C,Q; Compiled Lean; <code>f7e1fda; 93a5d98934c2</code>

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
QuotientGauss	weighted_transform_real_part. Under the finite-ring character assumptions in the source, the weighted real part of the quotient Kloosterman transform equals the squared Gauss-sum expression.	P,C,Q; Compiled Lean; f7e1fda; a8628dc031cf
AffineHHObstruction	concrete_mixed_level_displacement. The displayed mixed difference of four concrete rational hhLevel values equals 1040.	P,C,Q; Compiled Lean; f7e1fda; 9e3274b7f220
CompositeCompletion	zero_frequency_is_nonunit. In a nontrivial ZMod ring, zero belongs to the set of nonunits.	P,C,Q; Compiled Lean; f7e1fda; af643e3021ee
DeterminantCoordinates	concrete_expanded_tuple_sign_change. The two displayed expanded determinant tuples have signs +1 and -1 respectively.	P,C,Q; Compiled Lean; f7e1fda; aafce8f549d2

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
UnitSupported Completion	double_unit_supported_completion. For two functions vanishing on nonunits, the product of their unit-frequency moments satisfies the stated Gauss-normalized completion identity.	P,C,Q; Compiled Lean; f7e1fda; 7d84fa747313
PrimeCofactorIdentity	source_prime_bracket_cofactor. For an admissible prime/cofactor representation with the original quotient cap and unit condition, the source prime bracket has the explicit Mobius-logarithm factorization.	P,C,Q; Compiled Lean; f7e1fda; a31895518a60
ShortDivisorComplement	at_one. At argument one, the capped divisor kernel satisfies the displayed complementary short-divisor identity, for arbitrary natural caps.	P,C,Q; Compiled Lean; f7e1fda; 25f38fcb6f84
SquarefreeLcmCoefficient	squarefree_lcm_coefficient_unit. For squarefree P coprime to N and r dividing P, the unit-masked lcm/totient sum equals the explicit finite Euler product divided by r.	P,C,Q; Compiled Lean; f7e1fda; f791b4be0f73

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
ThreeAdicPrimePairing	actual_principal_partition. Under the recorded prime, ordering, coprimality and cap assumptions, the actual principal sum splits exactly into entropy, paired, singleton and geometric-face terms.	P,C,Q; Compiled Lean; f7e1fda; b3c22b714566
PrimeSemiprimeSwitch	actual_admissible_switch. For the specified admissible prime-to-semiprime switch, the two source brackets satisfy an exact logarithmic difference identity retaining both harmonic kernels.	P,C,Q; Compiled Lean; f7e1fda; 21462365bb2f
HarmonicKernelVariation	actual_kernel_abs_variation. For positive front and arguments, ordered cofactors and prime complements beyond Q, harmonic-kernel variation is bounded by the two explicit finite logarithm/totient sums.	P,C,Q; Compiled Lean; f7e1fda; 7bc1ee92e0e9
LeastMissingPrime Margin	large_prime_margin. For every natural p at least 13, the displayed harmonic-minus-logarithm expression is at least 1/144; primality is not required.	P,C,Q; Compiled Lean; f7e1fda; e1dbd4f8a68b

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
EulerAnchor	canonical_least_missing_prime_margin. For nonzero even N and the stated lower bound on the twin constant, the singular series exceeds the logarithm of the least missing odd prime by at least 1/144.	P,C,Q; Compiled Lean; f7e1fda; 88bbbb4d8e49
FourFormRoots	zero_mem_actualRoots. Zero is a root of the actual four-form product over ZMod, with the source parameters retained.	P,C,Q; Compiled Lean; f7e1fda; 49cbf93fd8eb
PowersetMoment	weighted_moment. For nonnegative finite subset weights, the first weighted powerset moment factors as a product times the sum of $h/(1+h)$ contributions.	P,C,Q; Compiled Lean; f7e1fda; 971f351decef
SelbergFourForms	weight_zero_outside. If every element of P is at least one, a divisor subset outside the defined Selberg support has zero weight.	P,C,Q; Compiled Lean; f7e1fda; b1b658aa92f8

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
FourFormTruncation	primeSupport_mono. The prime support at cutoff y is contained in that at cutoff z whenever y is at most z .	P,C,Q; Compiled Lean; f7e1fda; 577ff585207e
FourFormCollisionLoss	primeBase_positive. The defined collision base is strictly positive at every prime.	P,C,Q; Compiled Lean; f7e1fda; 000cade7cac3
SeparatedTypeII	xi_bound. The defined finite coefficient ξ has absolute value at most one for every finite E and natural v .	P,C,Q; Compiled Lean; f7e1fda; 4513f3323bbf
SeparatedTypeIIPrice	weighted_one_is_adversarial. The separated bilinear sum with constant weight one equals the defined adversarial sum.	P,C,Q; Compiled Lean; f7e1fda; dd971712c281

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
SeparatedTypeIICount	units_inclusion. For nonzero H , the count of elements coprime to H equals its Mobius divisor inclusion-exclusion sum.	P,C,Q; Compiled Lean; f7e1fda; 55ac598030d5
SeparatedTypeIILower	omitted_ratio_lower_R5. Under the explicit conductor, interval, unit-count and three front/density guards, the omitted-to-unit ratio is at least $\text{density}(H^*\text{ell})/(16^*\text{ell})$.	P,C,Q; Compiled Lean; f7e1fda; 5676b1d3be3a
DoubleExtraction Arithmetic	witnesses_distinct. The two constructed witnesses of an admissible Cell are distinct.	P,C,Q; Compiled Lean; f7e1fda; be85b2d59e5a
DividedFourForms	slopeProduct_eq. The product of the divided-form slopes equals minus e times the anchor times the cube of the modulus.	P,C,Q; Compiled Lean; f7e1fda; 613dba073eb1

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
DividedRootCounts	witness1_q_cast_ne_zero. For an admissible Cell, q is nonzero modulo the second constructed witness.	P,C,Q; Compiled Lean; f7e1fda; f7c4a4e4f2ca
DividedSelbergBridge	switch_weighted_moment. For an admissible Cell and nonsaturated local root counts, the switched powerset logarithmic moment equals its Euler product times the local-density sum.	P,C,Q; Compiled Lean; f7e1fda; 2b5f1edaaab6
RankCalibrationFace	three_small_primes_face_excludes_physicalMask. Three distinct prime divisors whose squares are at most the front exclude the defined arithmetic mask when their product divides the conductor-weighted argument.	P,C,Q; Compiled Lean; f7e1fda; 8bd774d17ae5
RankCalibrationPrice	weighted_rank_price_decomposition. The finite aggregate rank quantity decomposes exactly through an intermediate set into the remaining quantity and two price terms.	P,C,Q; Compiled Lean; f7e1fda; ce00b2b3d416

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
RankCalibration Arithmetic	squarefree_subdivisor_reconstruction. If d and d' are square-free, k divides d and k' divides d' , equality $d*k=d'*k'$ forces equality of both pairs.	P,C,Q; Compiled Lean; f7e1fda; 3ac4ec121db2
RankCalibrationUnitLoss	small_prime_dvd_small_conductor. Either designated prime c or r divides the defined small conductor whenever it is at most the cutoff R .	P,C,Q; Compiled Lean; f7e1fda; 2a3eaafd2e74
RankCalibrationEuler	weighted_unit_inclusion. For nonzero K , a finite sum over units modulo K equals its Mobius-weighted divisor inclusion-exclusion sum.	P,C,Q; Compiled Lean; f7e1fda; 30449aa225e9
RankCalibration Estimator	normalized_density_error_bound. For A at most J and positive ideal denominator, the difference of normalized A/J and A/ideal is bounded by the denominator discrepancy divided by ideal.	P,C,Q; Compiled Lean; f7e1fda; 2d175fb99479

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
TerminalPrimeExtraction	terminal_factorization . For n at least two, the constructed co-factor multiplied by the terminal prime equals n .	P,C,Q; Compiled Lean; f7e1fda; e7e6e458143b
BalancedResourceSwitch	three_strata . Every factor tuple falls into one of the three stated conductor ranges, separated by B and the square comparison with N .	P,C,Q; Compiled Lean; f7e1fda; 310338aef835
SignedHyperbolicCRT	unpack_encode . For an admissible resource cell, unpacking the signed encoding recovers the original encoding.	P,C,Q; Compiled Lean; f7e1fda; 69d0e23f36b1
NonSSBracketSwitch	theta_coordinate_reindex . The finite theta-bracket sum on the defined admissible domain equals its factor-coordinate reindexing.	P,C,Q; Compiled Lean; f7e1fda; 8d4cc04aa370

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
RankTwoHarmonic	unordered_pair_moment . The ordered half of a symmetric weighted pair sum equals one half of the sum of the squared total and the diagonal square sum.	P,C,Q; Compiled Lean; f7e1fda; 06cbe631518b
OddBonferroniArithmetic	truncated_choose_zero . For every truncation order, the alternating binomial sum at zero equals one.	P,C,Q; Compiled Lean; f7e1fda; e5348cf2301c
LeastFactorComposite	square_quotient . For prime p , the least-prime quotient of p squared equals p .	P,C,Q; Compiled Lean; f7e1fda; 4c114ae1b730
SwitchedSelbergWeight	totientDensity_properties . For even N , the totient density at every prime in the switched support is strictly between zero and one.	P,C,Q; Compiled Lean; f7e1fda; c0a640773636

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
PhysicalComposite Subtraction	physicalRaw_theta_difference . The defined raw arithmetic mass equals its prime mass plus the proper-prime-power difference.	P,C,Q; Compiled Lean; f7e1fda; 3d6ef57fd5fc
CompositeAPConductor	totient_product_kernel . For a finite set of primes, the Selberg product of totient densities equals the reciprocal totient of the prime product.	P,C,Q; Compiled Lean; f7e1fda; 04809ca5369c
SwitchedIncidence Estimator	switchedIncidence_reference_price . For even N and positive cutoff, the prime incidence minus its reference is bounded by the constructed main terms, both absolute remainders and the retained large-composite subtraction.	P,C,Q; Compiled Lean; f7e1fda; 7ea6cc850698
FriablePhysicalPrefix	smooth_large_has_actual_band_divisor . A Y -smooth integer n at least D has a divisor in the defined divisor band when $D > 1$ and $Y > 0$.	P,C,Q; Compiled Lean; f7e1fda; 4286a04e155a

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
FriableEulerRankin	tau_prime_pow . At a prime power p^k , the divisor-counting function τ equals $k+1$.	P,C,Q; Compiled Lean; f7e1fda; df9ddafb7b49
FriableKernelEnvelope	totientInverseSum_mono . The finite reciprocal-totient sum is monotone in its natural cutoff.	P,C,Q; Compiled Lean; f7e1fda; 80051a67fa9d
FriablePrimeHarmonic	pseries_tsum_le_one_add_inv . For positive σ , the nonnegative p -series with exponent $1+\sigma$ is at most $1+1/\sigma$.	P,C,Q; Compiled Lean; f7e1fda; 53df26ec4a58
FriableTotientEnvelope	weightedInverseTotient_multiplicative . The defined weighted inverse-totient arithmetic function is multiplicative.	P,C,Q; Compiled Lean; f7e1fda; 14419032ada2

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
FriablePhysicalPayment	uniqueResources1_subset_smoothInterval. The second set of unique smooth resources is contained in the defined smooth interval.	P,C,Q; Compiled Lean; f7e1fda; b4b108b42406
FriablePhysicalDemand	actual_theta_demand_abs_le_seven_log_cube. Under the recorded structural support and front guards with log N at least one, the theta bracket has absolute value at most $7(\log N)^3$.	P,C,Q; Compiled Lean; f7e1fda; b02cb0aced0d
FriableDemand Aggregation	friableRank_support. Membership in the defined friable rank implies structural support and smoothness of at least one of its two resources.	P,C,Q; Compiled Lean; f7e1fda; b564591ccb21
FriableSourceGeometry	source_u_pos. Under the defined source-onset condition, sourceU is positive.	P,C,Q; Compiled Lean; f7e1fda; bc249b736e46

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
FriableSourceBudget	source_rank_front_le_two_N. Under the source-onset condition, the product of the defined three source fronts is at most 2N.	P,C,Q; Compiled Lean; f7e1fda; 9dd811fe823f
EpsteinKernel22	boundary_deficit_bounds. For positive a and u, the defined boundary deficit lies between zero and $a^2/(2u^2)$.	P,C,Q; Compiled Lean; f7e1fda; b465bf53b124
EpsteinFinite22	finiteShiftIntegral_formula. For positive y and nonzero integer m, the finite shifted integral equals the stated weight-scaled endpoint sum.	P,C,Q; Compiled Lean; f7e1fda; fb6205d1d88b
EpsteinUnfold22	unfoldFullRealThreeHalves_natAbs. For positive y and nonzero integer m, the infinite shifted integral equals $2/(m ^2 \sqrt{y})$.	P,C,Q; Compiled Lean; f7e1fda; 79384d3502ab

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
EpsteinTail22	continuousOn_truncationEnvelopeReal. The defined real truncation envelope is continuous on the parameter domain where its second coordinate exceeds the fixed cutoff q.	P,C,Q; Compiled Lean; f7e1fda; 5f9d3f180df2
GammaPrerequisites22	Gamma_strip_exponential_bound. The actual Gamma function has norm at most $2 \exp(-\pi \text{Im } s /4)$ throughout the closed strip $1 \leq \text{Re } s \leq 2$.	P,C,Q; Compiled Lean; f7e1fda; 9f5e5fe14d18
GammaDerivative22	Gamma_derivative_strip_exponential_bound. The actual derivative of Gamma has norm at most $19 \exp(-\pi \text{Im } s /4)$ throughout the same closed strip.	P,C,Q; Compiled Lean; f7e1fda; 026ced6097c7
GammaPsiCore22	betaDifference_eq_integral. For $\text{Re } z > 0$ and real $w > 0$, the beta-integral difference equals the integral of the defined difference integrand on (0,1).	P,C,Q; Compiled Lean; f7e1fda; 450962be9526

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
GammaPsiBetaLimit22	gammaPsi_eq_beta_integral. For $\text{Re } z > 0$, the actual logarithmic derivative of Gamma equals minus Euler's constant plus its beta-integrand integral on (0,1).	P,C,Q; Compiled Lean; f7e1fda; b4adf3ac71dd
GammaPsiIntegral22	gammaPsi_eq_exp_integral. For $\text{Re } z > 0$, the logarithmic derivative of Gamma equals minus Euler's constant plus the defined exponential integral over positive reals.	P,C,Q; Compiled Lean; f7e1fda; 3f254cdfeee3
GammaBoxBounds22	gammaBoxRadius_continuousAt. The defined four-parameter Gamma-box radius is jointly continuous at every point whose Y coordinate is positive.	P,C,Q; Compiled Lean; f7e1fda; 4874192b1c2c
GammaContour Component22	gammaContourEnvelope_continuous. The defined two-parameter Gamma contour envelope is continuous.	P,C,Q; Compiled Lean; f7e1fda; 891454d4714e

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
ZetaEulerDirect22	contourZeta_ne_zero_on_right . The actual Riemann zeta function is nonzero for $\text{Re } s > 1$.	P,C,Q; partial row PASS; f7e1fda; 0d790ed67706
ZetaReflection22	contourZeta_logDeriv_reflection . On $-1 < \text{Re } s < 0$, the zeta logarithmic derivative equals the contour-chi logarithmic derivative minus the reflected zeta logarithmic derivative.	P,C,Q; partial row PASS; f7e1fda; cc39bd76e008
GammaPsiDuplication22	gammaPsi_duplication . For $\text{Re } z > 0$, the Gamma logarithmic derivative satisfies $\psi(z) + \psi(z+1/2) = 2\psi(2z) - 2\log 2$.	P,C,Q; partial row PASS; f7e1fda; 0491177c6cfc
GammaPsiReflection22	gammaPsi_shift_reflection . On $-1 < \text{Re } s < 0$, the indicated shifted Gamma-logarithmic-derivative difference equals $1/s$ minus the cotangent term.	P,C,Q; partial row PASS; f7e1fda; a931bbceb530

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
ContourChiPsi22	contourChi_logDeriv_eq_pair_integral . On $-1 < \text{Re } s < 0$, the contour-chi logarithmic derivative equals $\log \pi$ plus Euler's constant, $1/s$ and the actual paired integral.	P,C,Q; partial row PASS; f7e1fda; 30c44bb1e376
ThermalProjectionEnvelope22	thermalContinuousProjection_error_le . For $a > 0$ and natural N, M , the full-versus-partial circle projection error is bounded by the constructed projectionErrorEnvelope.	P,C,Q; partial row PASS; f7e1fda; 77cdab8bed77
ThermalProjectionIdentity22	thermalContinuousProjection_eq_vonMangoldt_sum . For $a > 0$ and natural N , the continuous circle projection equals the genuine von Mangoldt additive coefficient, including prime powers.	P,C,Q; Compiled Lean; f7e1fda; 8bb1eb5ec6ab
DiscreteThermalProjection22	sampled_trueCircle_eq_coefficient . For $N \leq M$ and $K > \max(N, 2M-N)$, the finite sampled projector equals the genuine coefficient for every real damping parameter a .	P,C,Q; Compiled Lean; f7e1fda; f24c702b0a6c

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
RationalLogQuantization22	quantizedLog_precision . For $2 \leq p \leq 1000000000$, the fixed rational log constructor has absolute error at most $1/\text{gridScale}$.	P,C,Q; Compiled Lean; f7e1fda; dfabcabf39ec
QuantizedLambdaEnvelope22	coefficient_error_bound . For $N \leq 1000000000$, the genuine coefficient differs from the scaled canonical integer coefficient by at most the explicit errorEnvelope.	P,C,Q; Compiled Lean; f7e1fda; 526354f4b960
FiniteFieldProjection22	fixed_A32_projection . Over a prime field larger than 2^{27} , a root of order 2^{27} gives the fixed canonical integer coefficient modulo that prime.	P,C,Q; Compiled Lean; f7e1fda; 2b1cc267ce78
ThermalGammaMellinInverse22	real_exp_eq_Gamma_integral . For real $x > 0$, $\exp(-x)$ equals the normalized actual Gamma inversion integral on $\text{Re } s = 2$.	P,C,Q; Compiled Lean; f7e1fda; daad8b5d5fb1

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
ConcreteNTTRoots22	primitive_3892314113 . The specified bank root modulo the certified prime 3892314113 has exact multiplicative order 2^{27} .	P,C,Q; Compiled Lean; f7e1fda; 16c414a5d2f0
ConcreteNTTA32Projection22	projection_3892314113 . The specified finite-field projector modulo 3892314113 returns the canonical integer coefficient at $N = 1000000000$.	P,C,Q; Compiled Lean; f7e1fda; cefc07197945
AngularMellinBorder22	angularBorder_one_ne_zero . For $a > 0$ and natural $N > 0$, the explicit angular boundary term at $q=1$ is nonzero.	P,C,Q; Compiled Lean; f7e1fda; 601a3360ff92
ComplexGammaMellinLocal22	complexGammaInverse_real . The defined complex Gamma inverse agrees with $\exp(-x)$ on every positive real x .	P,C,Q; partial row PASS; f7e1fda; e54cac5b2ab3

Module	Selected theorem and informal statement	Axioms; credit; commit; source SHA
ComplexGammaMellinHolomorphy22	<code>complex_exp_eq_Gamma_integral</code> . For $\text{Re } w > 0$, $\exp(-w)$ equals the normalized actual Gamma integral with principal complex powers.	P,C,Q; Compiled Lean; <code>f7e1fda; b8b69c16cdbe</code>
ComplexGammaMellinTail22	<code>exists_local_uniform_Gamma_tail</code> . For $\text{Re } w > 0$ and $H \geq 0$, a constructed neighborhood has Gamma tails bounded by twice the center radius uniformly for all cutoffs $T \geq H$.	P,C,Q; Compiled Lean; <code>f7e1fda; e5ba99611a06</code>
ComplexGammaMellinExpTailBridge22	<code>complex_exp_truncation_error_le</code> . For $\text{Re } w > 0$ and $H \geq 0$, the truncated Gamma inverse approximates $\exp(-w)$ within the explicit Gamma-tail radius.	P,C,Q; Compiled Lean; <code>f7e1fda; 4e804a6c6782</code>
ComplexGammaMellinLambda22	<code>lambdaMellinThermal_eq_integral</code> . For $\text{Re } w > 0$, the genuine infinite von Mangoldt exponential series equals its normalized Dirichlet-series/Gamma integral; the norm-summable interchange is derived.	P,C,Q; Compiled Lean; <code>f7e1fda; 4a94415678e4</code>

A.2 Existing finite-arithmetic obstruction groups

The following groups summarize existing negative declarations, support exclusions and retained correction terms. Each has its complete source hypotheses in the pinned Lean file and its recorded Judge credit in the electronic catalogue. The grouping is editorial; it introduces no new mathematical result.[A-ARITHMETIC-OBSTRUCTIONS]

Group; status	Guarded statement	What it does not show
Parity detector retains composites; Compiled Lean	Three distinct prime factors give $\text{oddWeight}=1$ while their product is composite; the explicit 101*103*107 example also has no prime factor at most 100.[A-OBSTRUCTION-PARITY]	This concerns the defined parity detector, not every sieve or the Goldbach coefficient.
Negative four-site block; Compiled Lean	The constructed closed four-site block is negative for every value of its displayed kernel parameter; the literal arithmetic block is negative as well.[A-OBSTRUCTION-MULTIFIBRE]	It refutes positivity of that block, not all possible kernels or arithmetic correlations.
Group; status	Guarded statement	What it does not show
Independent affine forms cannot produce a nonzero constant; Compiled Lean	Over the source field, independent affine forms cannot have an identity $f^*R + g^*S = N$ at every pair of arguments when N is nonzero. The rational and real specializations are recorded.[A-OBSTRUCTION-AFFINE]	The result concerns a global identity for the specified two affine forms; it does not forbid identities restricted to an admissible arithmetic set.
Expanded determinant signs vary; Compiled Lean	Two explicit expanded determinant tuples have opposite signs, +1 and -1.[A-OBSTRUCTION-SIGN]	No global sign for all determinant-coordinate tuples follows from the common coordinate construction.

Group; status	Guarded statement	What it does not show
Composite-modulus completion retains a nonunit defect; Compiled Lean	The exact finite character completion splits into its unit-frequency term plus nonunitDefect; zero is a nonunit in a nontrivial ZMod ring. Even for functions supported on units, the recorded defect formula is (q-kappa)*the defined arithmetic moment.[A-OBSTRUCTION-COMPLETION]	This records the correction term, not a claim that it is nonzero in every specialization.
Separated Type-II arithmetic and prime weights have different scope; Compiled Lean	Under the complete source guards, the adversarial arithmetic sum equals density times omittedCount. Incorporating ell into the calibrated modulus makes the bilinear term zero; with the stated candidate front the prime-weight bilinear term is also zero. The normalized omitted-ratio lower bound retains all three front/density guards R4a,R4b,R4c.[A-OBSTRUCTION-TYPEII]	An adversarial bounded arithmetic weight cannot be substituted for the actual prime weight; this does not refute a stronger weighted Type-II estimate.

Group; status	Guarded statement	What it does not show
Local collision factors remain in four-form bounds; Compiled Lean	The actual four-form and divided-form Euler lower bounds explicitly retain collisionLoss or switchCollisionLoss. Their truncated G bounds additionally retain ordering, coprimality, non-saturation, logarithmic-cutoff and prime-log-sum hypotheses.[A-OBSTRUCTION-COLLISION]	These are bounds with explicit factors and guards, not a proof that all collision losses must be large or prevent a sufficient estimate.
Three-small-prime rank faces exclude the arithmetic mask; Compiled Lean	Under the stated distinct-prime, small-square and divisibility assumptions, the three-prime face has beta=0 and is outside the arithmetic mask. With front at least 529 and 1771 dividing the conductor-weighted argument, the canonical profile and both its prime and raw-weight summands are nonpositive. The principal rank-price expression is nonpositive when its mass is nonnegative and t>1.[A-OBSTRUCTION-RANK]	The conclusion applies to these specified faces and canonical profiles; it does not prove a sign for the full reference gap.

Group; status	Guarded statement	What it does not show
Nonminimal prime-divisor switch terms are not positive; Compiled Lean	For the defined composite least-factor domain and its unit assumptions, switching on a prime divisor other than minFac gives a nonpositive odd-Bonferroni weight and a zero rough indicator.[A-OBSTRUCTION-LEASTFACTOR]	The sign is for the displayed switched summands, not a universal estimate for the complete composite correction.
Nonunit progression and resource support is excluded; Compiled Lean	On the specified admissible composite-progression domain, a nonunit modulus cannot satisfy the indicated congruence or divide the switched candidate; the corresponding prime progression sum vanishes. For an admissible ResourceCell, a divisor noncoprime to N cannot divide either resource, and a divisor containing the anchor cannot divide resource0.[A-OBSTRUCTION-NONUNIT-SUPPORT]	These exact support exclusions do not supply the required aggregate error budget.

A.3 Saved finite falsifiers in the historical analytic corpus

The following are existing finite Judge verdicts, not freshly evaluated banks or unconditional source-onset theorems. The stated claim is the one refuted within its saved bank. Entries reporting only no counterexample in a window are excluded.[A-FINITE-FALSIFIERS] Round 7 additionally records the original tuple $k = 7$: the proposed two-axis twists vanish while the native centered row is 5/6. This refutes that proposed connection at the recorded tuple only.[A-FINITE-NATIVE-CONNECTION]

Round	Claim refuted in its saved finite bank	Existing Judge status
14	every admissible parent has a neighbor in $H=2$ [A-FINITE-FALSIFIER-14-coverage.ERROR_FALSIFIER.0]	REFUTED_NEW_ZERO_F ULL_H_FAMILY
14	the complete expanded fibre P_all has universal coverage[A-FINITE-FALSIFIER-14-coverage.ERROR_FALSIFIER.1]	REFUTED_NEW_ZERO_F RONT_PREFIX
14	p-only shifts 2,4,6 cover the complete CRT105 cut[A-FINITE-FALSIFIER-14-complement.ERROR_FALSIFIER.0]	REFUTED_NEW_COMPLE TE_C1_ZERO_CAPACITY
14	H2 shifts on both large axes cover the complete mod35 cut[A-FINITE-FALSIFIER-14-complement.ERROR_FALSIFIER.1]	REFUTED_NEW_COMPLE TE_C2_ZERO_CAPACITY
Round	Claim refuted in its saved finite bank	Existing Judge status
15	replace the structural beta mask exactly by its unit density because d141 is small[A-FINITE-FALSIFIER-15-incidence.ERROR_FALSIFIER.0]	REFUTED_NEW_COMPLE TE_GAMMA_NONZERO
15	count each cofactor label as distinct parent capacity[A-FINITE-FALSIFIER-15-fusion.ERROR_FALSIFIER.0]	REFUTED_NEW_UNION_ MULTIPLICITY
15	all source-principal signs can be identified with the actual finite pair signs[A-FINITE-FALSIFIER-15-fusion.ERROR_FALSIFIER.1]	REFUTED_NEW_REAL_P AIR_SIGN
15	a six-distinct-unit-factor split with a large $q > a$ can lie below N in this finite bank[A-FINITE-FALSIFIER-15-fusion.ERROR_FALSIFIER.3]	REFUTED_NEW_RANK_M INIMUM
Round	Claim refuted in its saved finite bank	Existing Judge status
16	the uniform unit reference is exactly centered for Type I at divisor3 of the candidate[A-FINITE-FALSIFIER-16-typei.ERROR_FALSIFIER.0]	REFUTED_NEW_TYPEI3_ NONZERO_DRIFT
16	the observed least-prime capacity supplies enough once-only resources for the entire selected core family[A-FINITE-FALSIFIER-16-capacity.ERROR_FALSIFIER.0]	REFUTED_NEW_FINITE _FULL_PRINCIPAL_DE FICIT
16	erase $\Lambda(e)$ on all prime cores in the selected bracket[A-FINITE-FALSIFIER-16-capacity.ERROR_FALSIFIER.1]	REFUTED_NEW_PRIME_ CORE_LAMBDA_TERM
16	reuse the same e3 arithmetic capacity for each positive core[A-FINITE-FALSIFIER-16-capacity.ERROR_FALSIFIER.2]	REFUTED_NEW_RESOUR CE_REUSE

A.4 Algebraic corpus inventory

The accepted algebraic proof snapshot is **e5c4e119**; its existing Judge batch is **6f7d181a** and root acceptance is **3d5f0ff4**. The saved audit lists 233 theorem/lemma axiom entries, 204 complete frozen headers and 25 imported modules. The electronic inventory contains 356 named rows:

those 233 theorems/lemmas, 120 other named declarations and 3 constructors. Definitions and constructors are not added to the theorem/lemma count.[ALG-INVENTORY-SCOPE] The full inventory is supplied as `paper/algebraic_lean_inventory.csv` and `paper/algebraic_inventory.json`; the table below is a compact module catalogue followed by a *representative* theorem excerpt. The complete CSV records every row’s module, exact name, informal reading, source line and hash, source header, observed axiom list, credit and proof pin. Copied full source modules and original frozen headers are retained in `paper/algebraic_sources/`; this catalogue does not replace their typeclass, index, inequality, signed-function or certificate hypotheses.

In the tables, A3 means the existing print contains exactly `propext`, `Classical.choice`, `Quot.sound`; A2 means `propext`, `Quot.sound`; A0 means an explicitly empty print. P means `propext` alone. NR means no separately reprinted entry in the selected artifacts; it does not mean no axioms. CL denotes a compiled theorem/lemma, D a definition/type, and C a constructor in an accepted module. CL-C marks the conditional corollary whose retained hypothesis is discussed in Appendix B. All rows use accepted proof snapshot `e5c4e119`. Module names below omit the common `AlgebraicGoldbach.` prefix; `Calibration.` is retained. Exact full names are in the CSV/JSON.

Module	CL	Other	Scope of accepted module
<code>AlgebraicGoldbach[ALG-MODULE-01]</code>	0	0	Import collection for the accepted auxiliary modules; the frozen open target is excluded.
<code>BertrandFamily[ALG-MODULE-02]</code>	11	6	Actual rational Bertrand polynomial family, prime truth, and the $N \geq 24$ common-zero obstruction.
<code>CoprimeDivisor[ALG-MODULE-03]</code>	12	5	Actual rational coprime-divisor equations are equivalent to at most one prime coordinate differing from one.
<code>CoprimeKaryCountermodel[ALG-MODULE-04]</code>	6	2	An actual Boolean common zero at $N=4$ excludes a certificate for each $k \geq 2$ in this exact class.
<code>CoprimeKaryIdeal[ALG-MODULE-05]</code>	10	11	Equality of the actual rational arithmetic k-ary divisor ideal and the prime k-subset ideal.
<code>CoprimeKaryModels[ALG-MODULE-06]</code>	1	0	Arbitrary rational models are classified by fewer than k prime coordinates differing from one.
<code>FactorCoverage[ALG-MODULE-07]</code>	16	7	Proper-divisor coverage at horizon $M \leq N^2$ is equivalent to pins on primes with $p^2 \leq M$.
<code>LiteralCube[ALG-MODULE-08]</code>	17	23	Actual Boolean/coarse/ordered-g cube and a strict finite weighted covering obstruction.
Module	CL	Other	Scope of accepted module
<code>LowerBound.BooleanAnnihilation[ALG-MODULE-09]</code>	1	0	Boolean-equation multiples have zero aggregated coefficient image, without asserting ordinary polynomial zero.
<code>LowerBound.BooleanCoefficientDegree[ALG-MODULE-10]</code>	2	0	Aggregated coefficients vanish on supports larger than ordinary total degree.
<code>LowerBound.BooleanCoefficients[ALG-MODULE-11]</code>	4	3	Ordinary exponent-support aggregation and compatibility with genuine variable multiplication.
<code>LowerBound.BooleanCountBridge[ALG-MODULE-12]</code>	1	0	The genuine ordinary count polynomial maps to the actual count multiplier.
<code>LowerBound.BooleanLattice[ALG-MODULE-13]</code>	11	5	Rational Boolean-lattice adjoints, commutator, and strict low-layer raising injectivity.
<code>LowerBound.BooleanMultiplierRigidity[ALG-MODULE-14]</code>	2	1	Top-layer rigidity for the actual count multiplier, arbitrary rational parameter and signed functions.
<code>LowerBound.CountVariableCommutation[ALG-MODULE-15]</code>	1	0	The actual variable and count actions commute on finite types for arbitrary rational parameters.
<code>PinnedControl[ALG-MODULE-16]</code>	11	2	Circular coordinate-pinning calibration and its certificate equivalence; multiplier and summand degrees are distinct.

Module	CL	Other	Scope of accepted module
ProperBertrand[ALG-MODULE-17]	11	4	Interval-only family with $m \geq 2$, every-coordinate Boolean freedom, prime truth, and $N \geq 24$ obstruction.
SignDefinite[ALG-MODULE-18]	5	3	Canonical rational coefficients of one sign preserve every supplied prime-supported common zero.
Soundness[ALG-MODULE-19]	22	12	Ordinary polynomial certificate soundness, ordered Goldbach count including loops, and explicit faithful-field bounds.
UniformCountermodel[ALG-MODULE-20]	14	4	A Boolean coarse/Bertrand common zero for every $N \geq 24$, with composite coordinates allowed.
UniformNonpinning[ALG-MODULE-21]	2	0	The uniform coarse model differs from the prime indicator and selects composite nine.
Calibration.C1[ALG-MODULE-22]	26	12	Full-count calibration, prime support, unordered pairs including loops, and the exact independence threshold.
Calibration.Growth[ALG-MODULE-23]	5	0	The required ordinary-Q static-NS degree is unbounded along even N , without asserting existence of certificates.
Calibration.LowerBound[ALG-MODULE-24]	31	14	Universal ordinary-Q static-NS lower bound for the supplied full-count certificate, not an upper construction.
Module	CL	Other	Scope of accepted module
Calibration.PCRestriction[ALG-MODULE-25]	11	9	Genuine ordinary-Q PC restriction preserves rules and line degree; the lower corollary retains an external hypothesis.

A.5 Representative algebraic theorem excerpt

This selection is not the exhaustive theorem inventory. The electronic rows retain all actual audited names and their exact source headers. Displayed names omit only the common project prefix described above.

Declaration	Informal reading (source hypotheses retained)	Ax.	Level
certificate_excludes_common_zero[ALG-REP-01]	A supplied identity excludes a supplied common zero of all its generators.	A3	CL
certificate_implies_goldbach[ALG-REP-02]	A supplied certificate and prime-truth premises imply existence of the corresponding prime pair.	A3	CL
finiteFieldPolicy_g_nonzero_iff[ALG-REP-03]	The explicit field policy makes prime-point nonzero equivalent to positive integer count.	A3	CL
bit_boolean[ALG-REP-04]	The model's integer bit is idempotent.	P	CL
uniform_bertrand_countermodel[ALG-REP-05]	For every $N \geq 24$ the explicit bit model satisfies Booleanity, coarse constraints, intervals and $g=0$.	A3	CL
uniform_model_selects_composite_nine[ALG-REP-06]	For $N \geq 24$ the model selects nine whereas the prime indicator does not.	A3	CL
bertrandFamily_no_certificate[ALG-REP-07]	For $N \geq 24$ this exact coarse/Bertrand family has no supplied certificate identity.	A3	CL
pinned_certificate_iff_goldbach[ALG-REP-08]	Existence of a pinned-control certificate is equivalent to positive ordered count.	A3	CL

Declaration	Informal reading (source hypotheses retained)	Ax.	Level
pinned_multiplier_degree_le_one[ALG-REP-09]	The explicit pinned multipliers have ordinary degree at most one.	A3	CL
pinned_summand_degree_le_two[ALG-REP-10]	Each explicit pinned multiplier-generator summand has degree at most two.	A3	CL
FactorCoverage.coverage_iff_prime_pins[ALG-REP-11]	Under $M \leq N^2$ coverage is equivalent to prime pins for $p^2 \leq M$.	A3	CL
FactorCoverage.horizonFamily_no_certificate[ALG-REP-12]	The explicit smaller-horizon hypotheses exclude a certificate for this combined family.	A3	CL
ProperBertrand.family_does_not_pin_any_coordinate[ALG-REP-13]	For every coordinate the family has a Boolean model taking each of zero and one there.	A3	CL
ProperBertrand.family_no_certificate[ALG-REP-14]	For $N \geq 24$ the proper interval-only family has no certificate identity.	A3	CL
SignDefinite.sign_definite_strengthening_has_no_certificate[ALG-REP-15]	A supplied prime-supported Boolean/g common zero survives the same-sign strengthening and excludes its certificate.	A3	CL
CoprimeDivisor.family_iff_at_most_one_omitted_prime[ALG-REP-16]	The rational coprime-divisor equations hold iff at most one prime coordinate differs from one.	A3	CL

Declaration	Informal reading (source hypotheses retained)	Ax.	Level
CoprimeDivisor.family_does_not_pin_any_coordinate[ALG-REP-17]	For every coordinate the family has a Boolean model taking each of zero and one there.	A3	CL
CoprimeKaryIdeal.arithmeticIdeal_eq_primeIdeal[ALG-REP-18]	The actual rational arithmetic k-ary ideal equals the prime k-subset ideal.	A3	CL
CoprimeKaryModels.family_iff_omitted_card_lt[ALG-REP-19]	For arbitrary rational points, the k-ary family holds iff fewer than k prime coordinates differ from one.	A3	CL
CoprimeKaryCountermodel.family_no_certificate_at_four[ALG-REP-20]	For every $k \geq 2$ this exact k-ary family has no certificate at $N=4$.	A3	CL
LiteralCube.weighted_common_zero[ALG-REP-21]	For even $N \geq 6$ strict total clause weight below cube size leaves a full family/Boolean/coarse/g common zero.	A3	CL
BooleanLattice.adjoint[ALG-REP-22]	The actual total raising and lowering actions are adjoint.	A3	CL
BooleanLattice.commutator[ALG-REP-23]	The actual commutator acts by variable count minus twice support size.	A3	CL
BooleanLattice.up_injective_below_middle[ALG-REP-24]	On a pure k layer with strict $2k < \text{card}$ sigma, a zero raising image forces the signed function to vanish.	A3	CL

Declaration	Informal reading (source hypotheses retained)	Ax.	Level
BooleanCoefficients.coefficients_X_mul[ALG-REP-25]	Genuine ordinary variable multiplication maps to the actual finite-support bit action.	A3	CL
BooleanCoefficients.coefficients_eq_zero_of_totalDegree_lt_card[ALG-REP-26]	Aggregated coefficients vanish when support size exceeds ordinary polynomial degree.	A3	CL
BooleanCoefficients.top_layer_vanishes_of_countMul_degree_le[ALG-REP-27]	With strict $2k < \text{card}$ sigma, function and count image zero above k force the function zero at and above k.	A3	CL
BooleanCoefficients.coefficients_count_mul[ALG-REP-28]	The genuine ordinary count-polynomial multiple maps to the actual count multiplier.	A3	CL
BooleanCoefficients.coefficients_boolean_mul[ALG-REP-29]	An ordinary Boolean-generator multiple has zero support-coefficient image, including infinite variable types.	A3	CL
BooleanCoefficients.bitMul_countMul[ALG-REP-30]	On finite types the actual bit and count actions commute for arbitrary rational parameter and signed functions.	A3	CL
Calibration.C1[ALG-REP-31]	For natural m, a full D_m pseudo-model exists iff $m \leq \pi(N) - r(N)$, including loop pairs.	A3	CL
Calibration.D_full_count_pins[ALG-REP-32]	Every full-count D model agrees with the prime indicator within the horizon.	A3	CL

Declaration	Informal reading (source hypotheses retained)	Ax.	Level
LowerBound.Restriction.Dpi_multiplier_degree_lower_bound[ALG-REP-33]	The supplied ordinary-Q full-count certificate has count-multiplier degree at least π -r.	A3	CL
LowerBound.Restriction.Dpi_standard_degree_lower_bound[ALG-REP-34]	The count summand of a supplied ordinary-Q full-count certificate has degree at least π -r+1.	A3	CL
Calibration.Dpi_standard_degree_unbounded_on_even[ALG-REP-35]	Necessary count-summand degree for supplied full-count certificates is unbounded along even $N \geq 4$.	A3	CL
PCRestriction.dpi_refutation_restricts[ALG-REP-36]	A supplied D_{π} refutation yields a same-degree genuine knapsack refutation.	A3	CL

A.6 Complete finite degree rows (Evidence)

	N	π	r	α	NS $\mathbb{Q}/\mathbb{F}_p$	PC $\mathbb{F}_p$	PC $\mathbb{Q}$
10[ALG-DEGREE-N10]	4	2	2		3	3	3
12[ALG-DEGREE-N12]	5	1	4		5	3	3
14[ALG-DEGREE-N14]	6	2	4		5	3	3
16[ALG-DEGREE-N16]	6	2	4		5	3	3
18[ALG-DEGREE-N18]	7	2	5		6	4	4
20[ALG-DEGREE-N20]	8	2	6		7	4	4
22[ALG-DEGREE-N22]	8	3	5		6	4	4
24[ALG-DEGREE-N24]	9	3	6		7	4	4
26[ALG-DEGREE-N26]	9	3	6		7	4	4
28[ALG-DEGREE-N28]	9	2	7		8	5	5
30[ALG-DEGREE-N30]	10	3	7		8	5	5
32[ALG-DEGREE-N32]	11	2	9		10	6	—
34[ALG-DEGREE-N34]	11	4	7		8	5	—
36[ALG-DEGREE-N36]	11	4	7		8	5	—
38[ALG-DEGREE-N38]	12	2	10		11	—	—
40[ALG-DEGREE-N40]	12	3	9		10	—	—
42[ALG-DEGREE-N42]	13	4	9		10	—	—
44[ALG-DEGREE-N44]	14	3	11		12	—	—
46[ALG-DEGREE-N46]	14	4	10		11	—	—
48[ALG-DEGREE-N48]	15	5	10		11	—	—
50[ALG-DEGREE-N50]	15	4	11		12	—	—
52[ALG-DEGREE-N52]	15	3	12		13	—	—
54[ALG-DEGREE-N54]	16	5	11		12	—	—
56[ALG-DEGREE-N56]	16	3	13		14	—	—
58[ALG-DEGREE-N58]	16	4	12		13	—	—
60[ALG-DEGREE-N60]	17	6	11		12	—	—

Here $p = 1000000007$; a dash denotes an absent measurement, not a bound.

B Conditional and out-of-scope material

The predicate `KnapsackLowerBound` remains an explicit external hypothesis in the credited polynomial-calculus restriction corollary. The literature cited in the main text is not a Lean proof of this project predicate. No unconditional consequence requiring it is included. [R-PC]

The recorded `EulerAnchor.canonical_least_missing_prime_margin` also retains the quantitative premise $2541/4096 \leq \text{twinConstant}$, in addition to its nonzero even-input conditions. Its inventory row records a compiled implication under that premise. This report supplies no discharge of the quantitative premise and asserts no unconditional singular-series margin from that row. [analytic.declaration.EulerAnchor.GoldbachRound16.Anchor.canonical_least_missing_prime_margin]

Written analytic truncation and coefficient-connection sources without compile credit are reported with SOURCE status. The full signed balance estimate and native-program refinement remain open. Existing material depending on unproved project axioms or on physical analogies is excluded from the established-result inventory. Historical Horizon material lacking the required Judge credit is likewise excluded. The review checklist identifies these boundaries.[R-OUT-OF-SCOPE]

C AI assistance and responsibility

AI tools assisted artifact reading, drafting, table generation and review. They are not authors. Durand Serge is the sole named author and is responsible for the mathematical interpretation, accuracy, citations and submission. Agent checks and formal compilation do not imply external validation, peer review, journal acceptance or endorsement by a model provider. The reproducibility records identify the checks performed and their limits.

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